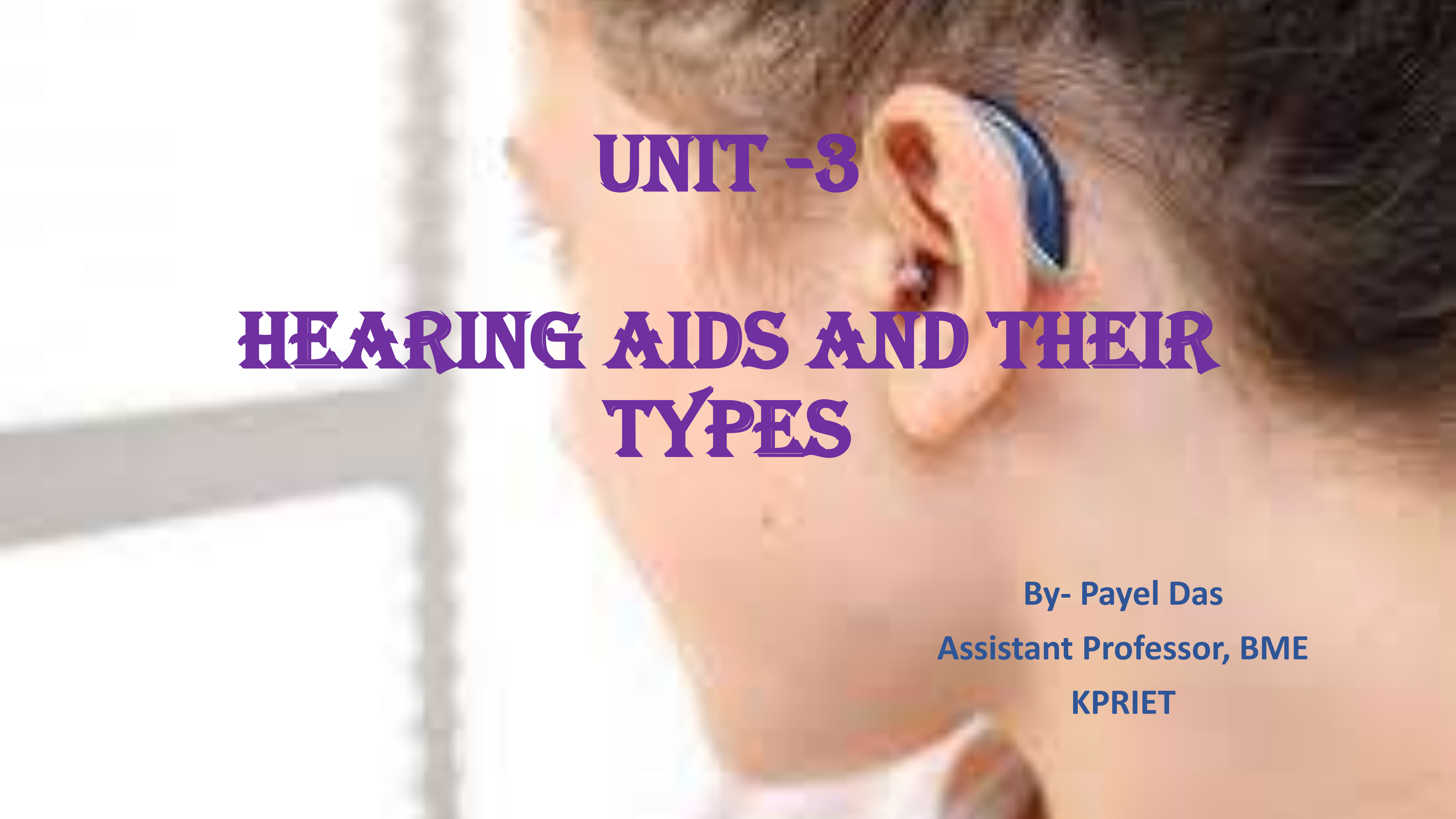

UNIT -3

HEARING AIDS AND THEIR TYPES

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KPRIET

Introduction

- Hearing aids are electronic devices designed to amplify sound for individuals with hearing loss.
- They play a crucial role in restoring auditory perception, improving communication, and enhancing overall quality of life.
- Hearing aids do not restore hearing but assist in making sounds clearer and more accessible.
- Captures sound → Processes it digitally → Amplifies and delivers it to the ear

Types of Hearing Aids

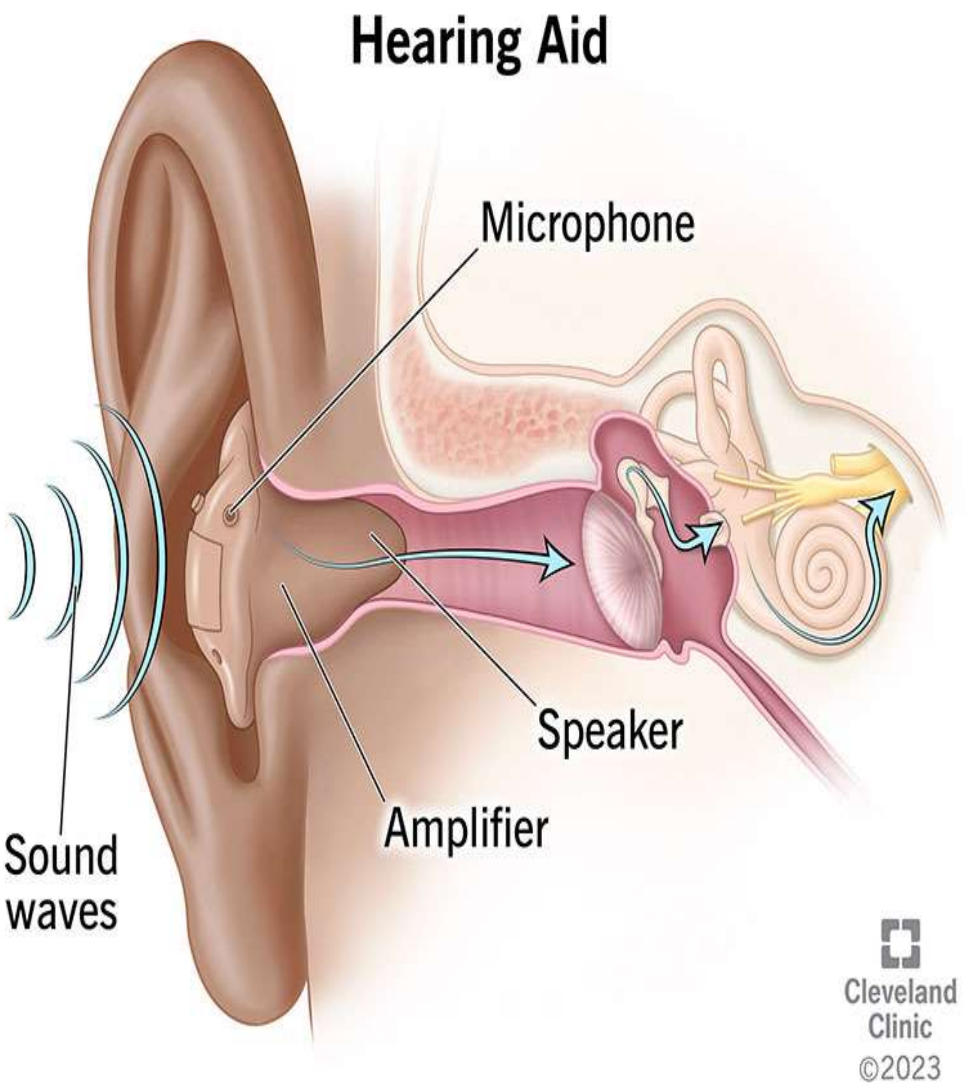
1. Behind-the-Ear (BTE)

2. In-the-Ear (ITE)

3. In-the-Canal (ITC) & Completely-in-the-Canal (CIC)

4. Receiver-in-Canal (RIC) or Receiver-in-the-Ear (RITE)

5. Bone-Anchored Hearing Aids (BAHA)


Hearing Aid
Microphone
Speaker
Amplifier
Sound
waves
Cleveland
Clinic
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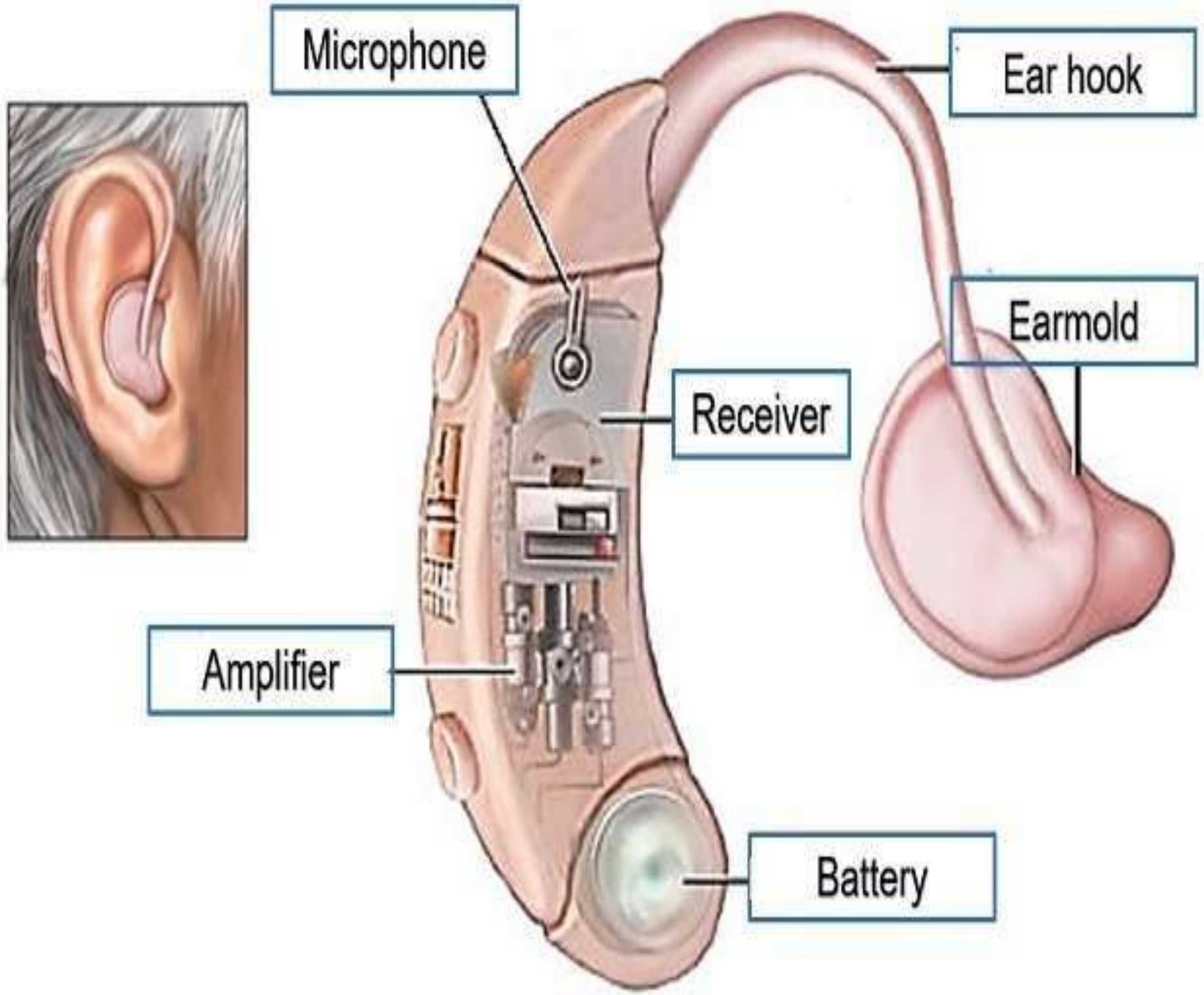

Microphone
Ear hook
Earmold
Receiver
Amplifier
Battery

Hearing Aids

In-the-ear (ITE) | In-the-canal (ITC) | Reciever-in-canal (RIC) | Behind-the-ear (BTE)

Behind-the-Ear (BTE) | Receiver-in-the-Canal (RIC) | In-the-Ear (ITE) | In-the-Canal (ITC) | Completely-in-the-Canal (CIC)

How Hearing Aids Work

1. Sound Capture: Microphone picks up sound waves.

2. Analog-to-Digital Conversion: Converts sound into digital signals.

3. Signal Processing: Digital signal processor (DSP) filters and amplifies sound.

4. Amplification: Sound is adjusted to match user's hearing needs.

5. Output Delivery: Receiver delivers the processed sound.

6. Battery Power: Device operates using rechargeable or disposable batteries.

Components of a Hearing Aid

1. Microphone: Captures sound.
2. Amplifier: Increases sound strength.
3. Digital Signal Processor (DSP): Adjusts sound clarity.
4. Receiver (Speaker): Delivers enhanced sound to the ear.
5. Battery: Powers the device.
6. Wireless Communication Module: Enables Bluetooth and telecoil connectivity.

Types of Hearing Aids

Hearing aids are classified based on design, placement, and functionality:

1. Behind-the-Ear (BTE)
2. In-the-Ear (ITE)
3. In-the-Canal (ITC) & Completely-in-the-Canal (CIC)
4. Receiver-in-Canal (RIC) / Receiver-in-the-Ear (RITE)
5. Bone-Anchored Hearing Aid (BAHA)

Behind-the-Ear (BTE) Hearing Aids

- Worn behind the ear with a tube connecting to an earmold.
- Suitable for mild to profound hearing loss.
- High amplification power, long battery life, durable.
- Available in mini-BTE versions for discretion.

In-the-Ear (ITE) Hearing Aids

- Custom-molded to fit the outer ear.
- Suitable for mild to severe hearing loss.
- Easier to handle, can include volume controls.
- Larger than ITC/CIC models but still discreet.

In-the-Canal (ITC) & Completely-in-the-Canal (CIC)

- ITC fits partly inside the ear canal, CIC fits deeper.
- Suitable for mild to moderate hearing loss.
- Cosmetically appealing, nearly invisible.
- Requires frequent maintenance due to earwax buildup.

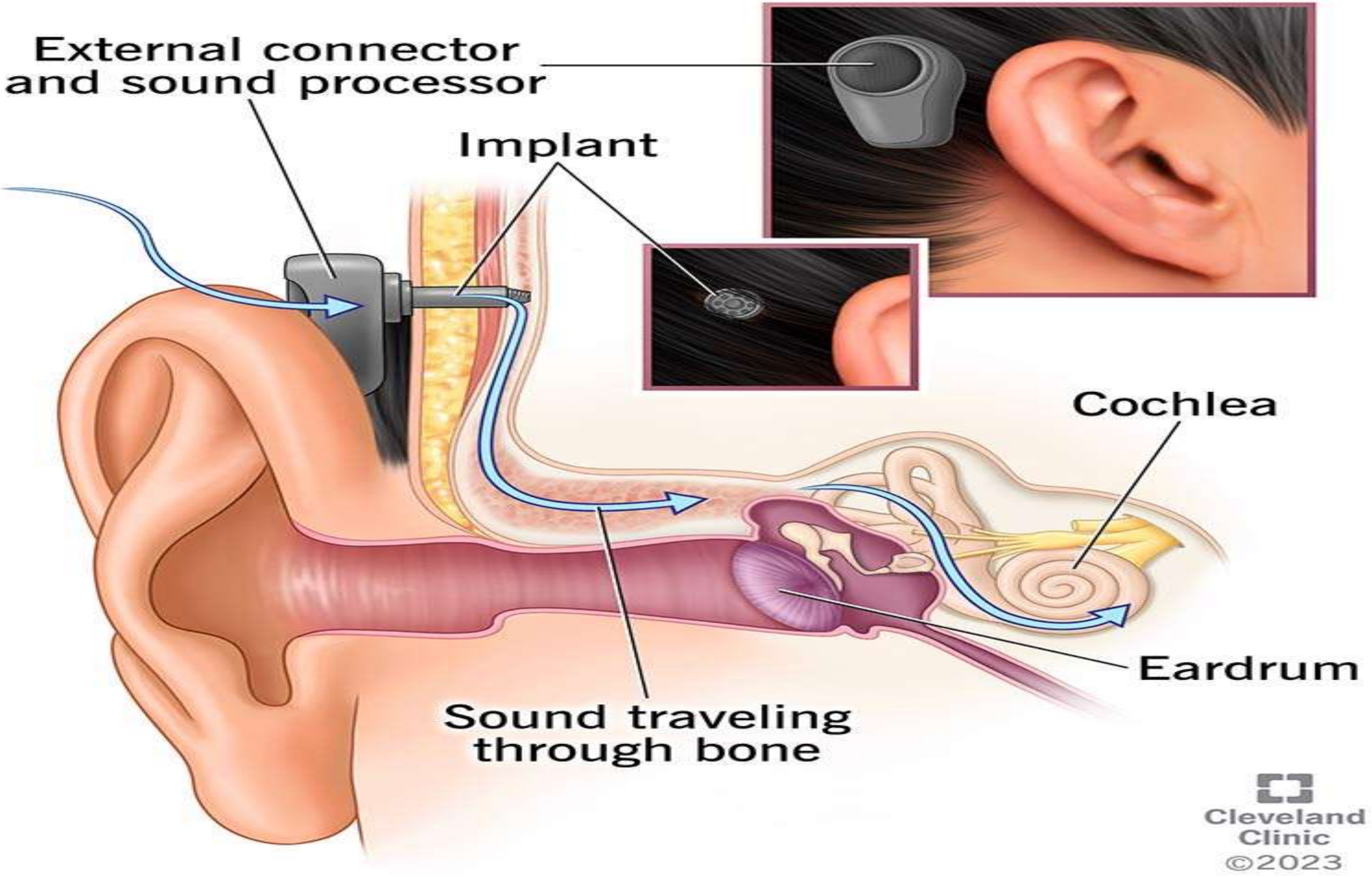
Receiver-in-Canal (RIC) / Receiver-in-the-Ear (RITE)

- Similar to BTE but with the receiver inside the ear canal.
- Suitable for mild to severe hearing loss.
- Provides clearer sound, less feedback.
- More discreet than traditional BTE models.

Bone-Anchored Hearing Aids (BAHA)

- Uses bone conduction instead of air conduction.
- Suitable for conductive hearing loss or single-sided deafness.
- Requires surgical implantation.
- Provides clear sound transmission through skull vibrations.

Bone Anchored Hearing Aid



Technological Advancements in Hearing Aids

- AI-powered hearing aids adapt to different environments.
- Bluetooth connectivity for wireless audio streaming.
- Rechargeable batteries for convenience.
- Noise reduction algorithms enhance speech clarity.

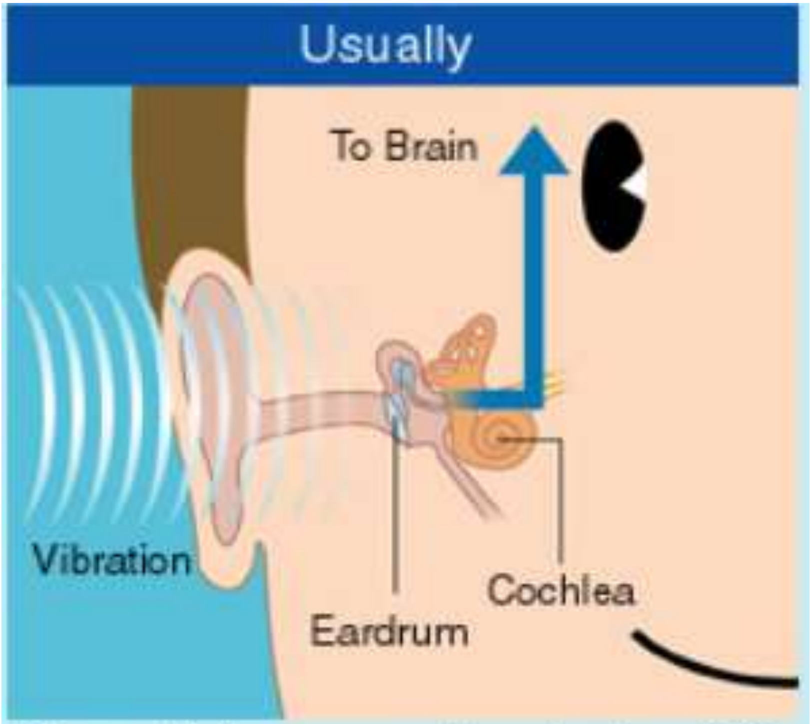
Type	Placement	Best for	Advantages	Disadvantages
BTE	Behind the ear, with tube & earmold	Mild to profound loss	High power, long battery life, Bluetooth	Visible, bulky
ITE	Custom-fit in outer ear	Mild to severe loss	Easy to handle, manual controls	More visible, wind noise
ITC/CIC	Partially (ITC) or fully (CIC) inside the canal	Mild to moderate loss	Discreet, natural sound	Short battery life, frequent maintenance
RIC/RITE	Behind ear, receiver in canal	Mild to severe loss	Clearer sound, less feedback	Delicate components, costly
BAHA	Implanted in the skull	Conductive or SSD loss	Bypasses damaged parts, natural sound	Requires surgery, risk of infection

How we usually hear

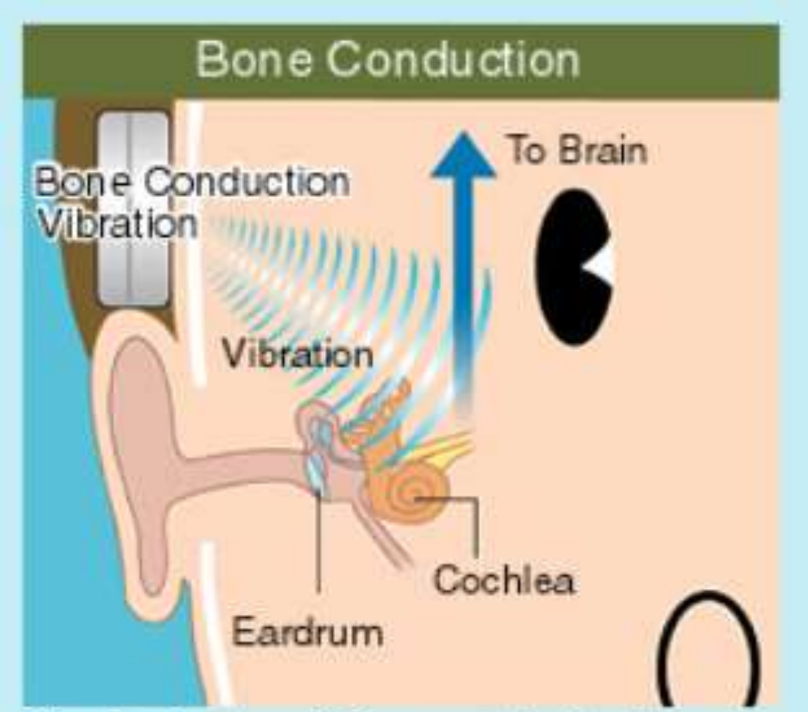
- Normal sound waves are actually tiny vibrations in the air. The vibrations travel through the air to our eardrums. The eardrums in turn vibrate, decoding these sound waves into a different type of vibrations that are received by the Cochlea, also known as the inner ear. The Cochlea is connected to our auditory nerve, which transmits the sounds to our brain.

How we hear with Bone conduction

- Bone Conduction bypasses the eardrums. In bone conduction listening, the bone conduction devices (such as headphones) perform the role of your eardrums. These devices decode sound waves and convert them into vibrations that can be received directly by the Cochlea so the eardrum is never involved. The “sound” reach the ears as vibrations through the bones (or skull) and skin.



We usually hear a sound by vibration of air.



The Basic Idea of Bone conduction is,
Bone Vibration makes us hearable.

Bone-Anchored Hearing Aids (BAHA)

- BAHA are implantable hearing devices designed to improve hearing in individuals with conductive hearing loss, mixed hearing loss, or single-sided deafness (SSD).

- BAHA utilizes bone conduction to directly stimulate the cochlea, bypassing the outer and middle ear.

- **Principle of Bone conduction -** Sound waves are transmitted as mechanical vibrations directly through the bones of the skull to reach the inner ear (cochlea), bypassing the external ear and middle ear structures.

Components of a BAHA System

A BAHA system typically consists of the following components:

1. **Titanium Implant**: A small titanium screw is surgically implanted into the skull, usually behind the ear. The implant integrates with the bone through osseointegration, providing a stable foundation.

2. **Abutment (or Magnetic Attachment)**: Connects the external sound processor to the implant. Some models use a percutaneous abutment (direct connection) while others use a magnetic coupling system.

3. **External Sound Processor**: Contains a microphone, digital signal processor (DSP), and amplifier. It captures sound, processes it, and converts it into mechanical vibrations that are transmitted to the skull.

Working Mechanism of BAHA

1. **Sound Capture**: The external processor picks up sound waves via a microphone.
2. **Signal Processing**: The sound processor filters, amplifies, and converts audio signals into mechanical vibrations.
3. **Bone Conduction Transmission**: Vibrations are transferred from the external processor to the skull through the implant.
4. **Cochlear Stimulation**: The vibrations reach the cochlea, stimulating the auditory nerve and enabling sound perception.

Indications for BAHA Implantation

BAHA is recommended for patients with:

- **Conductive Hearing Loss**: Conditions like chronic otitis media, atresia, or ossicular chain defects where sound cannot travel through the outer/middle ear.

- **Mixed Hearing Loss**: Combination of conductive and sensorineural hearing loss where air conduction devices are ineffective.

- **Single-Sided Deafness (SSD)**: When one ear is completely non-functional, BAHA transmits sound from the impaired side to the functional cochlea on the opposite side, improving spatial awareness and speech perception.

Surgical Procedure for BAHA Implantation

1. **Preoperative Assessment**: Audiological and medical evaluations are conducted to determine candidacy.

2. **Anesthesia**: Performed under local or general anesthesia.

3. **Implant Placement**: A small incision is made, and the titanium implant is drilled into the mastoid bone.

4. **Healing and Osseointegration**: The bone fuses with the implant over 3–6 months.

5. **Processor Fitting**: The external sound processor is attached and programmed for patient-specific hearing needs.

Advantages of BAHA

- **Bypasses Middle Ear Pathologies**: Ideal for patients with chronic ear infections or anatomical abnormalities.
- **Superior Sound Quality**: Direct bone conduction reduces distortion and improves speech clarity.
- **Comfortable and Discreet**: No need for an in-ear device, reducing irritation.
- **Effective for Single-Sided Deafness**: Enhances directional hearing and spatial awareness.

Technological Advances in BAHA

- **Magnetic Attachment Systems**: Eliminates the need for a skin-penetrating abutment, improving aesthetics and reducing infections.
- **Wireless Connectivity**: Integration with Bluetooth for direct streaming to smartphones and assistive devices.
- **AI-Driven Sound Processing**: Adaptive algorithms enhance speech perception in noisy environments.
- **3D-Printed Implants**: Custom-designed implants improve biocompatibility and performance.

SIGNAL PROCESSING AND DESIGN CONSIDERATIONS OF A COCHLEAR IMPLANT DEVICE

Cochlear Implants

- A neuroprosthetic device that restores partial hearing in individuals with severe sensorineural hearing loss.

- **Function:** Bypass damaged hair cells in the cochlea and provide electrical stimulation to the auditory nerve.

- **Key Difference from Hearing Aids:** Cochlear implants do not amplify sound; instead, they convert it into electrical signals.

Components of a Cochlear Implant

External Components

- **Microphone** – Captures sound from the environment.
- **Speech Processor** – Processes and converts sound into digital signals.
- **Transmitter Coil** – Sends processed signals wirelessly to the implanted device.

Internal Components

- **Receiver/Stimulator** – Converts digital signals into electrical impulses.
- **Electrode Array** – Stimulates different regions of the cochlea.

Signal Processing in Cochlear Implants

1. **Sound Acquisition** - Captures sound via an external microphone.
2. **Preprocessing & Noise Reduction** - Enhances speech clarity while reducing background noise.
3. **Feature Extraction** - Identifies key speech elements for improved intelligibility.
4. **Signal Compression** - Adjusts sound intensity levels to fit the patient's hearing range.
5. **Frequency-to-Channel Mapping** - Assigns sound frequencies to specific electrodes in the cochlea.
6. **Electrical Pulse Generation** - Converts processed signals into neural stimulation.

Sound Acquisition & Preprocessing

- Microphone captures sound waves**.**

- **Preprocessing Techniques:**
  - **Noise Reduction** – Eliminates background noise for better speech perception.
  - **Directional Microphones** – Focus on sounds coming from a specific direction.
  - **Automatic Gain Control (AGC)** – Adjusts volume levels dynamically.

Feature Extraction & Signal Compression

- **Feature Extraction:**
  - Identifies key auditory cues such as pitch, intensity, and formants.
  - Uses techniques like Mel-Frequency Cepstral Coefficients (MFCCs).
- **Signal Compression:**
  - Reduces the dynamic range of sound signals for better hearing comfort.
  - Ensures soft sounds are audible while loud sounds do not cause discomfort.

Frequency-to-Channel Mapping

- **Cochlea's Tonotopic Organization:**
  - Different frequencies stimulate specific cochlear regions.
- **Frequency Mapping Techniques:**
  - **Fixed Band Allocation** – Assigns a fixed frequency range to each electrode.
  - **Dynamic Allocation** – Adjusts electrode activation based on incoming sounds.

Electrical Pulse Generation

- **Biphasic Electrical Pulses:**
  - Ensures safety by preventing long-term electrode damage.
- **Controlled Pulse Rate & Amplitude:**
  - Modulates intensity and frequency for clear speech perception.
- **Stimulation Strategies:**
  - **Continuous Interleaved Sampling (CIS)** – Reduces electrode interference.
  - **Fine Structure Processing (FSP)** – Preserves fine details of speech.

Design Considerations for Cochlear Implants

- **Biocompatibility & Materials:**
  - Titanium and medical-grade silicone prevent immune rejection.
- **Power Efficiency:**
  - Low-power circuits ensure extended battery life.
- **Electrode Array Design:**
  - **Perimodiolar Electrodes** – Wrap around the cochlea for precise stimulation.
  - **Straight Electrodes** – Easier insertion with minimal cochlear damage.
- **Wireless Connectivity:**
  - Bluetooth-enabled devices allow direct smartphone integration.
- **Customizability:**
  - Individualized speech mapping based on user preferences.
- **Safety & Reliability:**
  - Hermetic sealing prevents moisture-related damage.

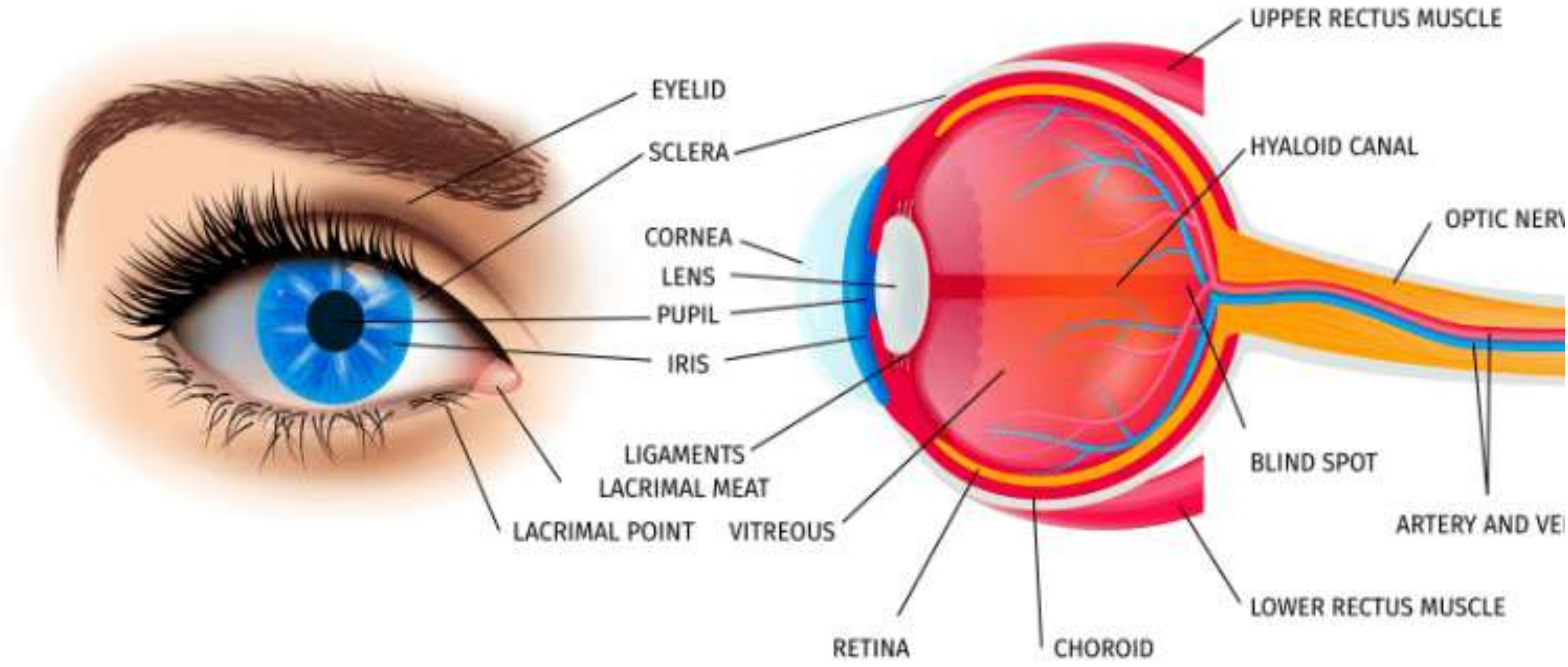
Advanced Signal Processing Strategies

- **Continuous Interleaved Sampling (CIS):**
  - Ensures non-overlapping pulses for better sound clarity.
- **Fine Structure Processing (FSP):**
  - Captures rapid fluctuations in sound waves.
- **Current Steering:**
  - Creates virtual channels to enhance spectral resolution.
- **Machine Learning & AI:**
  - Adapts processing strategies based on real-time auditory feedback.

CORNEA IMPLANT

Introduction

- The human eye is a complex sensory organ responsible for vision.
- It detects light, processes visual information, and sends signals to the brain for interpretation.
- The eye functions like a camera, focusing light to form images.
- Cornea must be clear, smooth and healthy for good vision. If it is scarred, swollen, or damaged, light is not focused properly into the eye. As a result, your vision is blurry


UPPER RECTUS MUSCLE
EYELID
SCLERA
HYALOID CANAL
CORNEA
LENS
PUPIL
IRIS
LIGAMENTS
LACRIMAL MEAT
LACRIMAL POINT
VITREOUS
BLIND SPOT
RETINA
CHOROID
LOWER RECTUS MUSCLE

- The cornea is the transparent front part of the eye that refracts light entering the eye to focus it on the retina.

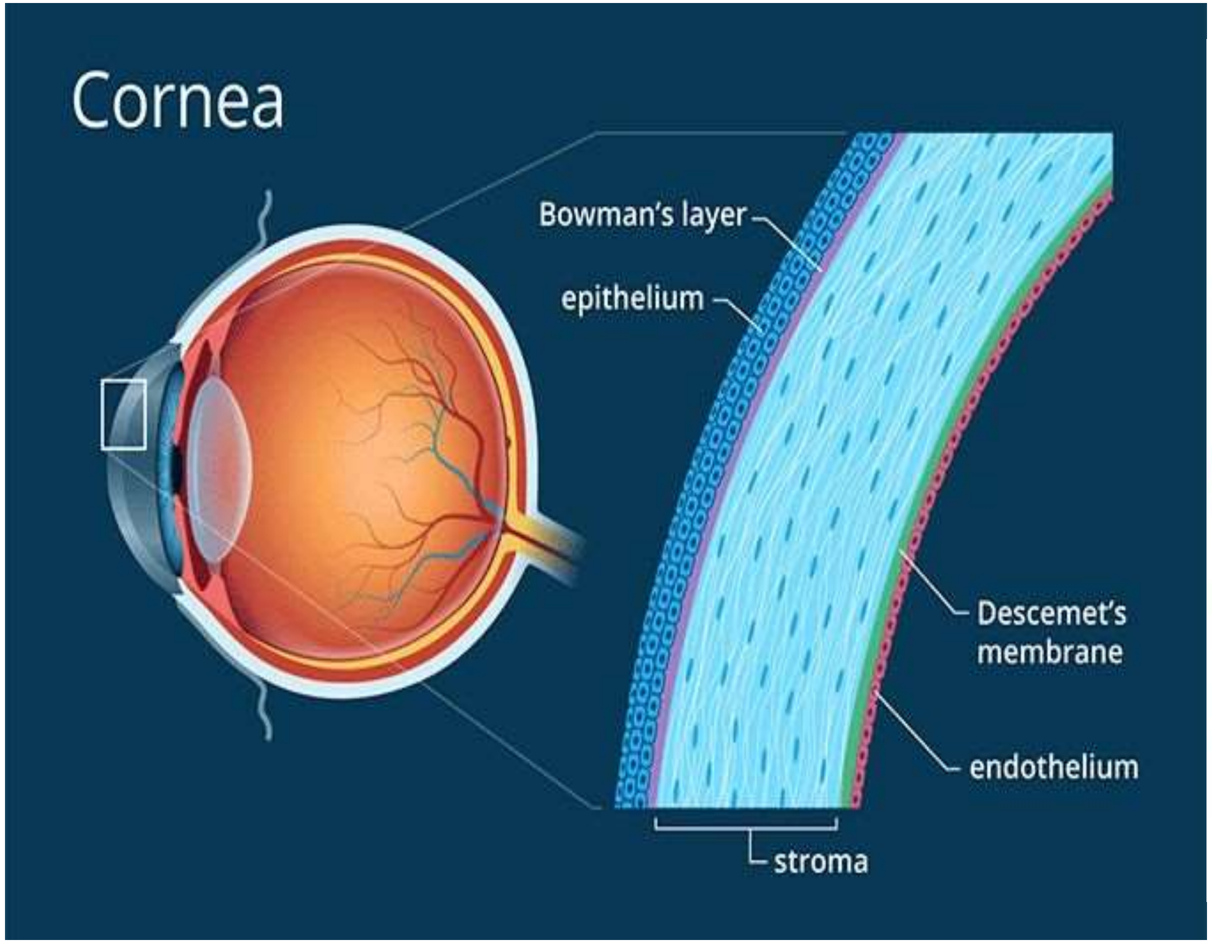
- Damage to the cornea due to injury, infection, or diseases such as keratoconus can lead to vision impairment or blindness.

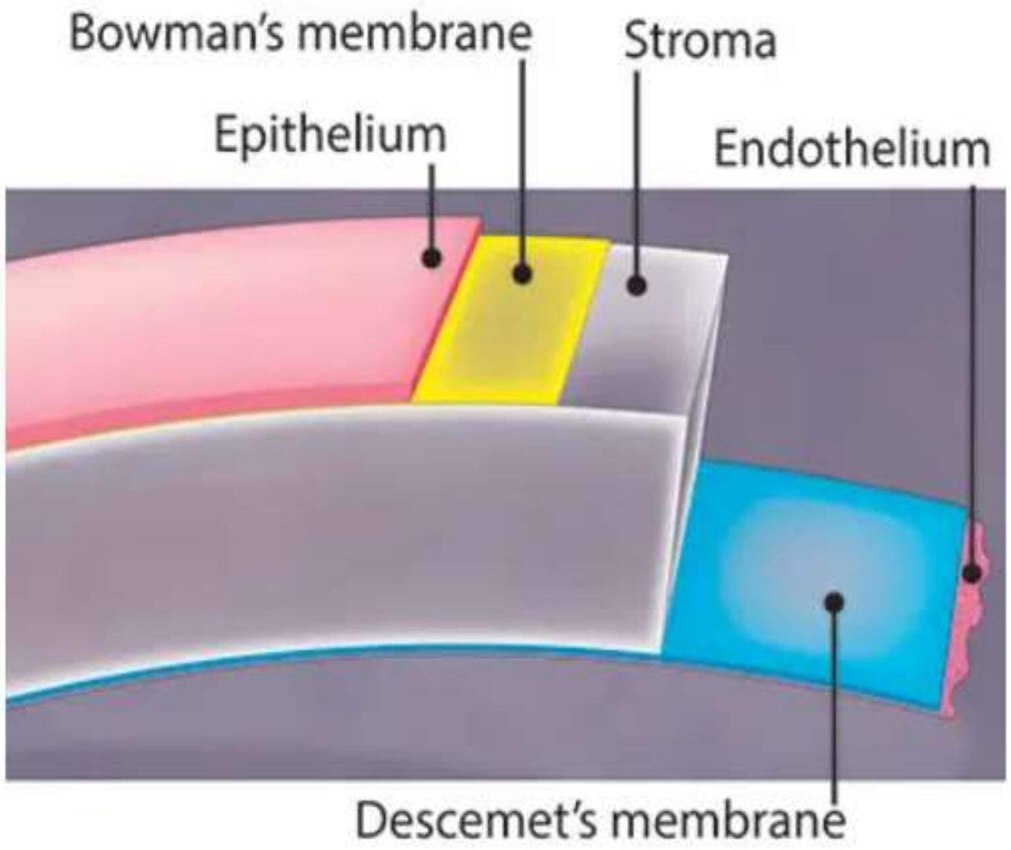
- Cornea implants, also known as corneal grafts or keratoprostheses, are used to replace damaged corneal tissue and restore vision.

Anatomy and Function of the Cornea

The cornea consists of five distinct layers:

1. **Epithelium** – The outermost layer that protects against foreign particles and absorbs nutrients.
2. **Bowman’s Layer** – A tough layer that provides structural integrity.
3. **Stroma** – The thickest layer, composed of collagen fibers, which gives the cornea its strength and transparency.
4. **Descemet’s Membrane** – A thin layer that supports the inner endothelium.
5. **Endothelium** – Maintains corneal hydration by regulating fluid exchange.


Cornea
Bowman's layer
epithelium
Descemet's membrane
endothelium
stroma


Bowman's membrane
Stroma
Epithelium
Endothelium
Descemet's membrane

Anatomy and Function of the Cornea

1. **Epithelium** – The outermost layer that protects against foreign particles, absorbs nutrients, and helps in oxygen exchange. It has a high regenerative capacity and heals quickly from minor injuries.
2. **Bowman’s Layer** – A tough, acellular layer composed of collagen fibers that provides structural integrity and acts as a protective barrier to underlying layers.
3. **Stroma** – The thickest layer, composed of precisely arranged collagen fibers and keratocytes, which provide the cornea with strength, transparency, and resistance to mechanical stress. It constitutes approximately 90% of the corneal thickness.
4. **Descemet’s Membrane** – A thin, strong, and elastic basement membrane that serves as a support for the inner endothelial layer. It thickens with age and provides structural reinforcement to the cornea.
5. **Endothelium** – A monolayer of hexagonal cells responsible for maintaining corneal hydration by regulating fluid exchange. These cells do not regenerate; damage to the endothelium can lead to corneal edema and vision loss.

Types of Cornea Implants

Cornea implants can be categorized into two main types:

1.Biological (Human Donor) Corneal Implants:

1. These involve transplanting a cornea from a human donor (cadaveric cornea).
2. The procedure is known as **penetrating keratoplasty (PKP)** when the entire cornea is replaced or **lamellar keratoplasty** when only a portion is replaced.
3. The most commonly used biological implant is derived from an eye bank.
4. Some patients may require **Descemet's Stripping Endothelial Keratoplasty (DSEK)** or **Descemet's Membrane Endothelial Keratoplasty (DMEK)** to replace only the endothelial layer.

2. Artificial (Synthetic) Corneal Implants:

1. Also called **keratoprosthesis (KPro)**, these implants are made of biocompatible materials such as PMMA (polymethyl methacrylate).
2. Used in cases where donor corneal transplantation is not possible due to severe damage or rejection risks.
3. Examples include **Boston Keratoprosthesis (Boston KPro)**, **AlphaCor**, and **Osteo-Odonto-Keratoprosthesis (OOKP)** (involves using a patient’s tooth as a base for the artificial cornea).

- Cornea implants can be categorized based on the extent of corneal tissue replacement. When the entire cornea is replaced with a healthy donor cornea, the procedure is called **penetrating keratoplasty (PKP)**.
- when only a portion of the cornea is replaced, it is known as **lamellar keratoplasty**. This approach preserves healthy corneal layers while selectively replacing the damaged portion.

Lamellar keratoplasty can be further divided into:

- **Deep Anterior Lamellar Keratoplasty (DALK):** Replaces the outer corneal layers, leaving the endothelium intact, useful for conditions like keratoconus.
- **Endothelial Keratoplasty (EK):** Targets the inner corneal layers, particularly the endothelium, and includes techniques such as Descemet's Stripping Endothelial Keratoplasty (DSEK) and Descemet's Membrane Endothelial Keratoplasty (DMEK), which are used for endothelial dysfunctions like Fuchs' dystrophy.

Procedure for Corneal Transplantation

1. Preoperative Evaluation:

1. Comprehensive eye examination, including visual acuity test, corneal topography, and pachymetry.
2. Donor screening and cross-matching for biological corneal transplant to reduce rejection.
3. Artificial corneal implant selection based on patient condition.

2. Surgical Techniques:

1. **Penetrating Keratoplasty (PKP):** Full-thickness corneal transplantation for advanced corneal diseases.
2. **Deep Anterior Lamellar Keratoplasty (DALK):** Selective replacement of corneal stroma while preserving healthy endothelium.
3. **Endothelial Keratoplasty (EK):** Used for endothelial dysfunction cases; includes DSEK and DMEK.
4. **Keratoprosthesis (KPro):** Artificial corneal implant for cases of repeated transplant failure or severe ocular surface disorders.

3.Postoperative Care:

1. Administration of antibiotic and steroid eye drops to prevent infection and immune rejection.
2. Regular follow-ups to monitor for complications such as graft rejection, infections, and corneal haze.
3. Vision rehabilitation through corrective lenses, glasses, or specialized contact lenses.

Design Considerations for Cornea Implants

1. Biocompatibility

- Implants must be **non-toxic** and not trigger an immune response.
- Common materials: **PMMA (Polymethyl methacrylate), collagen, and hydrogels**.

2. Optical Properties

- Transparency and clarity are essential.
- Minimal light scattering and **anti-reflective coatings** improve function.

3. Structural Stability & Durability

- Resistance to mechanical stress and wear.
- Hydrophilic or hydrophobic surface modifications for **moisture retention**.

4. Neural Connectivity

- In cases where implants interact with **optic nerve signals**, seamless neural integration is crucial.
- Electrodes must have a **high signal-to-noise ratio** for clear signal transmission.

5. Minimization of Post-Surgical Complications

- Risk of infections, **inflammation**, or implant rejection must be minimized.
- Bioactive coatings with **antimicrobial properties** help improve longevity.

Benefits of Corneal Implant Technology

- **Vision Restoration:** Replaces opaque or damaged tissue, allowing light to pass through clearly and improving vision.
- **Reduced Dependence on Glasses or Lenses:** Many patients achieve improved unaided vision after recovery.
- **Minimized Rejection Risk:** Especially in lamellar and endothelial keratoplasty where selective replacement preserves more native tissue.
- **Shorter Recovery Time:** Newer techniques like DMEK offer quicker visual recovery compared to traditional methods.
- **Life-Changing Impact:** Restoring vision significantly enhances quality of life, independence, and productivity.

Indications for Cornea Implantation

- Corneal scarring from infections (e.g., herpes simplex, fungal keratitis)
- Keratoconus (thinning of the cornea)
- Corneal dystrophies (e.g., Fuchs' dystrophy, lattice dystrophy)
- Chemical or traumatic injury to the cornea
- Corneal rejection from a previous transplant
- Corneal perforation due to severe ulcers
- Severe dry eye syndrome that leads to corneal damage

Retinal implants

- Retinal implants are **biomedical devices** designed to restore partial vision in patients suffering from **degenerative retinal diseases** like:

1. **Retinitis Pigmentosa (RP)**
2. **Age-related Macular Degeneration (AMD)**

In diseases like **RP** or **AMD**

- **Photoreceptors** degenerate and die.
- But inner retinal cells (like bipolar and ganglion cells) **may remain functional**.

- Retinal implants **bypass damaged photoreceptors** and electrically stimulate the remaining viable retinal cells to send visual information to the brain.

Anatomy of the Retina

The **retina** is a thin, light-sensitive tissue lining the **back of the eye**. It functions like the **film in a camera**, converting light into electrical signals that the brain can interpret as vision.

It consists of several layers of cells, including:

- **Photoreceptors (Rods and Cones)** – Convert light into electrical signals.
- **Bipolar Cells** – Transmit signals from photoreceptors to ganglion cells.
- **Ganglion Cells** – Send electrical impulses to the brain via the optic nerve.
- **Retinal Pigment Epithelium (RPE)** – Supports photoreceptors by supplying nutrients and removing waste.

Physiology of the Retina

- Light entering the eye is focused on the retina.
- **Photoreceptors** convert light into electrical signals through **hyperpolarization**.
- Signals travel through **bipolar cells** to **ganglion cells**, which generate **action potentials**.
- These signals are carried via the **optic nerve** to the **visual cortex** of the brain.
- Brain interprets the signals as visual images

Structure of Retinal Implants

Retinal implants are bioelectronic devices surgically implanted into or near the retina to restore partial vision in patients with photoreceptor degeneration (like Retinitis Pigmentosa or Age-Related Macular Degeneration).

Components:

- **External Camera**: Mounted on eyeglasses; captures images.
- **Image Processor**: Worn externally; converts images into digital signals.
- **Transmitter Coil**: Sends processed signals wirelessly to the implant.
- **Receiver Coil**: Implanted near the eye; receives signals.
- **Electrode Array**: Placed on (epiretinal), under (subretinal), or near (suprachoroidal) the retina; stimulates retinal neurons electrically.

Working Mechanism of Retinal Implants

1. **External Camera System (for some implants)**: A mini camera mounted on glasses captures the surrounding environment.
2. **Image Processing Unit**: Converts images into electrical signals and transmits them wirelessly to the implant.
3. **Microelectrode Array**: Implanted in the retina; receives signals and stimulates the appropriate retinal cells.
4. **Optic Nerve Transmission**: Electrical signals travel through the optic nerve to the brain, where they are interpreted as visual images.
5. Camera → Image Processor → Wireless Transmission → Retina → Optic Nerve → Brain

Function of Retinal Implants

1. Camera captures visual information.
2. Image processor simplifies and digitizes the image.
3. Signal is sent wirelessly to the implanted receiver.
4. Electrodes stimulate retinal cells (ganglion or bipolar).
5. Signals travel via the **optic nerve** to the **visual cortex**, where they are interpreted as vision.

Types of Retina Implants

1. Epiretinal Implants

Epiretinal implants are placed **on the inner surface of the retina**, particularly over the **ganglion cell layer**.

Working Mechanism:

These implants use an **external camera mounted on eyeglasses** to capture images. The images are processed by a wearable unit and then transmitted wirelessly to the implant. The electrodes in the implant **stimulate ganglion cells**, which send the signal through the **optic nerve** to the brain.

Advantages:

- Suitable for patients with complete photoreceptor loss.
- **Example Device:**
  Argus II (developed by Second Sight Medical Products, USA)

2. Subretinal Implants

Subretinal implants are placed **beneath the retina**, in the **photoreceptor layer**, between the **retina and the retinal pigment epithelium (RPE)**.

Working Mechanism:

These implants are equipped with **microphotodiodes** that respond to incident light directly, converting it into electrical signals. The stimulation is directed at the **bipolar cells**, mimicking the natural flow of visual signals.

Advantages:

- Utilizes natural eye movements and optics.
- Does not require an external camera.
- Offers more natural image perception.
- **Example Device:**
  Alpha AMS (developed by Retina Implant AG, Germany)

3. Suprachoroidal Implants

Suprachoroidal implants are placed in the **suprachoroidal space**, which lies between the **sclera** and the **choroid** layer, thus making the implant more external compared to the other types.

Working Mechanism

Images are captured via an **external camera**, processed, and transmitted to the implant, which **stimulates the retina through the choroid and RPE** layers indirectly.

Advantages

- **Minimally invasive** surgery.
- Lower risk of damaging retinal tissue.
- Easier to remove or replace.
- **Example Device**
  Suprachoroidal Prototype developed by Bionic Vision Australia (BVA)

Materials and Technology Used

- **Microelectrodes** – Made from biocompatible materials like platinum and silicon.
- **Wireless Communication** – Uses RF (radio frequency) technology for power and data transfer.
- **Flexible Substrates** – Ensures minimal damage and better integration with retinal tissue.

Advantages of Retina Implants

- Restores partial vision in blind individuals.
- Helps patients identify shapes, objects, and motion.
- Enhances mobility and independence.

PROSTHETIC EYE

Introduction and Definition

A **prosthetic eye**, also known as an **ocular prosthesis** or **artificial eye**, is a medical device that replaces a missing natural eye. While it does **not restore vision**, it plays a critical role in:

- Restoring **facial symmetry** and aesthetics.
- **Improving psychological well-being** of the patient.
- Supporting surrounding tissues and maintaining **orbital volume**.
- The device typically fits over an orbital implant placed during surgical procedures like **enucleation** or **evisceration**.

Purpose of a Prosthetic Eye

1. Cosmetic Appearance
   - Restores natural look by mimicking the real eye.
2. Socket Health
   - Prevents tissue shrinkage and maintains facial structure.
3. Psychosocial Benefits
   - Helps boost self-confidence and reduces social anxiety.
4. Eyelid Support
   - Maintains eyelid shape and function.

Clinical Background and Indications

Indications for Prosthetic Eye

- **Enucleation**: Complete removal of the eyeball due to tumors, trauma, or infection.
- **Evisceration**: Removal of eye contents, leaving the scleral shell intact.
- **Congenital anophthalmia**: Absence of the eye at birth.
- **Severe trauma**: Where vision cannot be salvaged.
- **Phthisis bulbi**: A shrunken, non-functional eye due to chronic disease.

These conditions may be caused by:

- Retinoblastoma (eye cancer in children)
- Penetrating injuries
- End-stage glaucoma or infections

Anatomy and Functional Considerations

- The **eye socket** must have a proper anatomical structure for fitting.
- The **extraocular muscles**, if preserved, can be reattached to a motility implant to allow limited prosthetic eye movement.
- Healthy **eyelids** and **tear secretion** are essential for comfort and natural appearance.

Materials Used

- **PMMA (Polymethyl Methacrylate)**: Most commonly used. Biocompatible, rigid, polishable, and lightweight.
- **Medical-grade Acrylics**: Used for scleral shell formation.
- **Silicone**: Used for orbital implants or liners (not commonly for the visible prosthesis).

- **Porous Materials**:
  - **Hydroxyapatite**: Allows tissue ingrowth, can be integrated with muscles for motility.
  - **Porous Polyethylene (e.g., Medpor)**: Lightweight and porous, also used for integration.

Types of Prosthetic Eyes

Type	Description	Pros	Cons
Stock Prosthesis	Pre-manufactured, selected by size and color	Quick fit, low cost	Poor customization
Custom Prosthesis	Individually designed using patient’s socket measurements and iris matching	Natural look, better fit, comfort, and motility	Time-consuming, expensive

Construction of a Prosthetic Eye

The construction of a prosthetic eye is a highly personalized process that typically involves the following steps:

1. Patient Evaluation

- The ocularist (a specialist who makes artificial eyes) examines the eye socket.
- The socket must be fully healed (after surgery or trauma) before the prosthesis is fitted.

2. Impression Molding

- A mold of the eye socket is made using soft impression material.
- This mold captures the exact shape and size of the socket to ensure a proper fit.

3. Wax Model Fitting

- A wax model (trial shape) is created and placed in the socket to check comfort and movement.
- Adjustments are made to ensure a natural look and fit.

4. Iris and Sclera Painting

- The iris (colored part of the eye) is hand-painted to match the natural eye.
- Details such as blood vessels on the sclera (white part) are added for realism.

5. Final Prosthesis Fabrication

- The final prosthesis is made using acrylic resin.
- The eye is cured, polished, and sterilized before insertion.

6. Fitting and Follow-up

- The prosthetic eye is gently inserted into the socket.
- Patients are taught how to care for and clean it.
- Follow-up appointments ensure comfort and cosmetic success.

Limitations of a Prosthetic Eye

1. **No Vision Restoration**
   A prosthetic eye is cosmetic and does not restore sight. The person remains blind in the affected eye.
2. **Limited Movement**
   Although some prostheses may move slightly with the natural eye, the movement is often limited and less natural.
3. **Need for Regular Maintenance**
   The prosthesis must be cleaned regularly to prevent irritation or infection. Annual polishing by an ocularist is also recommended.
4. **Changes in Eye Socket Over Time**
   The shape of the eye socket may change due to tissue remodeling, requiring refitting or replacement of the prosthetic eye.
5. **Risk of Discomfort or Irritation**
   If not fitted properly, the prosthetic eye can cause discomfort, dryness, or socket inflammation.
6. **Cost and Limited Accessibility**
   Custom-made prosthetic eyes can be expensive and may not be easily available in all regions, especially in rural or low-resource areas.

Why a Prosthetic Eye Does Not Restore Vision

A **prosthetic eye** is **purely cosmetic** and **structural**—it **does not have any optical, sensory, or neurological function.**

1. No Optical Components

Unlike an artificial lens in cataract surgery or a retinal implant, a prosthetic eye has:

- **No cornea**, **no retina**, **no optic nerve**.
- It does **not collect or transmit light**.
- It simply replaces the **external appearance** of the eyeball.

2. No Connection to the Brain

Vision requires:

- Light to enter the eye → focus on retina → convert to electrical signals → transmit to brain via the **optic nerve**.
- In people who need a prosthetic eye, this entire visual pathway is **damaged or absent**.
- So, even if you place an artificial globe that "looks like" an eye, there is **no signal transmission to the brain**.